



Biofeedback as a stress management tool: a systematic review

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Abstract

Inappropriate management of acute stress can negatively affect cognition and task performance. Frequently occurring acute stress encounters can lead to cardiovascular and immunity deficiencies, and psychological disorders such as depression, fatigue, and burnout. Biofeedback can be used as a non-invasive, passive, continuous method of managing stress in real time. A systematic review of biofeedback as a real-time stress management intervention for non-patients was conducted to identify literature between 2000 and 2017, yielding 17 studies evaluating physiological, psychological, and/or performance metrics. Participants represent convenience samples ($N=9$ studies) and deliberately selected samples, whose optimal performance under stress is critical for occupational success ($N=8$ studies). Various methods to collect data, display biofeedback, induce stress, and measure performance were reported. Overall, biofeedback is an effective intervention that can be used to reduce physiological and subjective stress, and enhance performance. This is especially true among professionals, whose job performance requires appropriate stress management.

Keywords Acute stress · Chronic stress · Stress management · Biofeedback · Performance

1 Background

Imagine being a volunteer at the Boston Marathon in 2013, and watching as the celebratory scene transforms into shock and chaos. The sounds of explosions fill your ears. People scatter to determine the cause of chaos and help those in need. Your palms start sweating, your pupils dilate, and your heart rate and breathing rate increase. Your body is preparing you to escape to safety as efficiently as possible. Jumbled thoughts ensue, but your concern is physically leaving danger to find safety. This is an illustration of the adaptive and beneficial effects of the short-term acute stress response initiated by the sympathetic nervous system (Allen 1983; Greenberg et al. 1980; Venables et al. 2004; Zhai et al. 2006).

Now imagine being a physician in one of Boston's hospitals, suddenly flooded with patients in need of urgent care, with no prior knowledge of the situation and no time to waste. Identical physiological cascades take place by way of sympathetic nervous system activation, characterizing the acute stress response. Acute stress is defined by its short-term, immediate, and transient physiological changes that occur when we are confronted with some form of perceived danger (Tanev et al. 2014). However, the chemical and endocrine imbalances intended to prepare the body to fight or flee advance over minutes as systemic cortisol release reaches the hippocampus, amygdala, and prefrontal cortex (LeBlanc 2009; McEwen 2006). As a victim attempting to escape to safety, the cognitive side effects of this acute stress endocrine cascade—including decision-making, attentional, and memory deficits—have little effect on the overall goal. However, to the physician, whose performance relies on optimal functioning of decision-making, memory recall, and cognitive control and determines the outcome of the patient, the cognitive alterations associated with cortisol release stimulated by an acute stressor are detrimental and need to be managed appropriately.

In addition to the detrimental effect of discrete stressors on cognition and performance, repeated exposure to stressors with little recovery time between events also presents

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a danger. Excessive levels and extended durations of acute stresses characterize chronic stress (McEwen 2006; Tanev et al. 2014). Burnout is a subset of prolonged or chronic stress that is extremely prevalent among helping professions, particularly among healthcare practitioners (Balch et al. 2009; Rosenstein 2012). Physician burnout is associated with a threat to patient safety, diminished quality of patient care, and an increase in medical errors and adverse events (Balch et al. 2009).

Managing stress within an optimal range and in moderation ensures the highest potential for peak performance (Allen 1983; Hupbach et al. 2012; Vine et al. 2013). The quantification of positive stress, or eustress, and detrimental stress, or distress, can be overlaid with the Yerkes-Dodson law to illustrate the curvilinear relationship between performance and stress (Fig. 1). The amount of stress required to achieve optimal performance varies by activity and individual, emphasizing the need to personalize stress management approaches.

The stress referred to in Fig. 1 can be physical, mental, or emotional (Greenberg et al. 1980). Emotional and cognitive/mental stress are of particular interest, because they specifically contribute to disruptions in decision-making, attentional, and memory processes (Cohen et al. 2015). Furthermore, cognitive stress can often have a more direct impact on performance (Sime 2007). In high-risk professions, poor performance is unacceptable. High-risk professions are characterized by frequent occurrences of high-workload situations managed by small groups operating within risky environments (Klampfer et al. 2001). Examples commonly include teams found in the military, air craft crews, surgical teams, police force, teams working in nuclear power plants, etc. When performance is affected, particularly in such high-risk professions, where physically and cognitively demanding acute stressors are commonplace and any error can have catastrophic consequences, understanding the contributors to performance under stress and developing mitigation strategies is necessary (Delahaij et al. 2011, 2017).

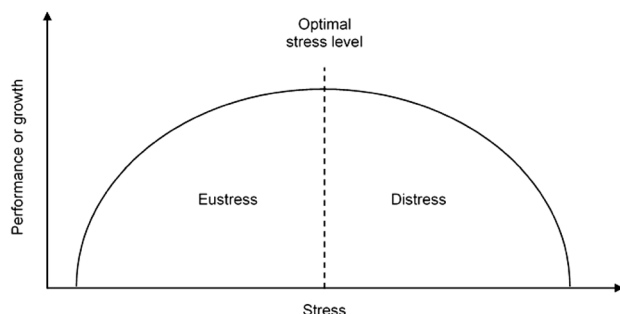


Fig. 1 Quantitative eustress–distress relationship, adapted from Allen (1983)

A comprehensive understanding of the interactions among physiological, cognitive, and performance variables occurring under stress has been outlined in the neurovisceral integration model (Thayer et al. 2009, 2012). This model is built on the basic empirical evidence that cardiovascular functioning, indexed by heart rate (HR) and heart rate variability (HRV), is mediated via dual-innervation of both the sympathetic nervous system and the parasympathetic nervous system. Though both branches typically achieve a dynamic balance in the resting state, humans become susceptible to physiological and cognitive disruptions—starting at the level of the prefrontal cortex (PFC) and ultimately influencing HRV and cognitive performance—when a static imbalance is reached. In the context of acute stress, static imbalance is initiated as a physiologically adaptive response to environmental demands, with potentially maladaptive cognitive consequences in the absence of appropriate behavioral adaptation. The neurovisceral integration model proposes that an overarching central nervous system (CNS) network controls psychophysiological attentional, emotional, and executive functioning resources. According to the model, associated neural structures—including the PFC, cingulate cortex, insula, amygdala, hypothalamus, and others—are responsible for coordination of behavioral adaptations. Thus, this broad and interactive CNS network enables flexibility and adaptability in the face of environmental demands. Maladaptive patterns and behaviors emerge when this network is uncoupled and not working appropriately (Thayer et al. 2009, 2012).

The neurovisceral integration model, along with empirical pharmacological and neuroimaging evidence (Thayer et al. 2012), supports evidence that cortical activity (namely PFC cortical activity) modulates associated cardiovascular function (namely HRV). The model further provides empirical evidence that performance on executive functioning (i.e., decision-making and memory processes) and attention tasks is affected by the same top–down process. Though stress can impair these executive functioning and attentional abilities and can negatively affect HRV, increasing resting levels of HRV as a means of stress management can improve cognitive functioning (Thayer et al. 2009). This can be attributed to the role of HRV in performance on tasks recruiting PFC-localized abilities (i.e., executive functioning and attention processes). Therefore, theory suggests, and evidence confirms, that enhancing HRV can improve PFC-localized cognitive abilities and in turn, improve performance on tasks relying on those abilities. In most experimental literature, HRV is measured as a dependent variable, but the neurovisceral integration model incorporates HRV as an independent variable, capable of active manipulation to produce a desired outcome or set of outcomes (Thayer et al. 2009, 2012). Introducing behavioral programs has been demonstrated to be an effective way to improve cognitive functioning and

performance by way of manipulating HRV (Thayer et al. 2009). One such behavioral intervention capable of directly addressing HRV is biofeedback.

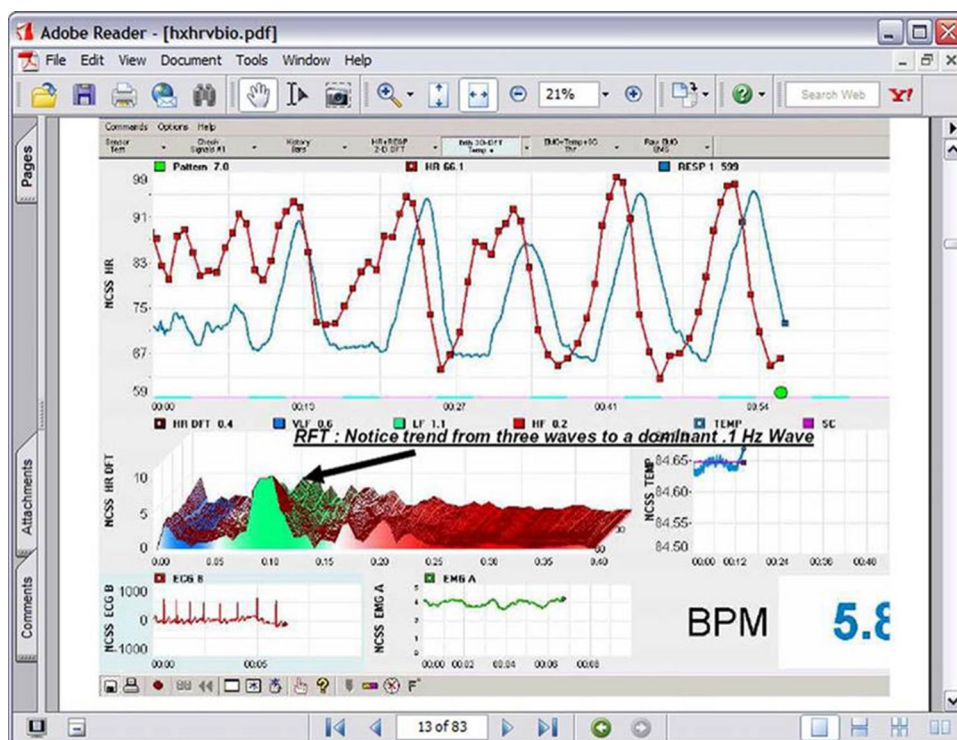
Biofeedback is a mind–body intervention that has the benefit of being unobtrusive, passive, and continuous. Unlike many mind–body interventions addressing stress (e.g., Lehrer 2007), biofeedback externalizes an individual's physiological state and allows the user to monitor changes in real time (Schwartz 2010). Thus, as the stress response unfolds, users can track its progression, and take steps to preemptively mitigate any detrimental changes. With training over time, these processes can become more automatic, with little or no reliance on the biofeedback instrument (Schwartz 2010). While biofeedback can measure a variety of psychophysiological variables and can deliver data back to the user through a variety of modalities, HRV biofeedback as a specific subtype of biofeedback is a standardized method with manuals, protocols, and guides to illustrate its uses (Eddie et al. 2015; Lehrer et al. 2000, 2013).

HRV biofeedback consists of monitoring HR oscillations using a visual display, and learning to voluntarily generate a simple sinusoidal curve on the display, representing cross-coherence, by actively altering one's breathing rate. Cross-coherence between systems considered to be oscillatory—those with repetitive temporal variation—occurs when two rhythmic systems, such as heart rate and breathing rate, match in frequency (McCraty et al. 2009). By changing the frequency of breaths, a HRV biofeedback user can induce HRV changes, blood pressure changes, activate the

baroreflex, and achieve coherence among the cardiovascular systems. Breathing at a rate of 9–24 breaths per minute, which translates to 0.15–0.4 Hz in frequency, is typical for most individuals, though this rate and its inverse, frequency, varies depending on the individual and the context. HR oscillations that correspond to this breathing rate are referred to as respiratory sinus arrhythmia (RSA), which has the same range in frequency (0.15–0.4 Hz) as the “high frequency” (HF) band of the HRV power spectrum. The HF/RSA index is associated with regulatory functions and parasympathetic tone via vagal nerve innervation of the heart (Lehrer et al. 2014).

The goal in HRV biofeedback training is for the user to arrive at the appropriate breathing rate that coincides with their RSA frequency, resulting in a 0° phase relationship, or optimal coherence/overlap, of the two sinusoidal curves representing: (1) breathing rate and (2) HF/RSA amplitudes (Fig. 2). Achieving this goal results in a coherent HRV waveform, where both traces are matched over time (refer to McCraty et al. 2009 for details on calculating the heart rhythm coherence ratio). In practice, this 0° phase relationship, or ideal HRV waveform, is observed as times when heart rate and breathing occur concurrently. In other words, optimal coherence is obtained when heart rate rises simultaneously as a user inhales, and falls simultaneously as a user exhales. Additional influences on overall coherence and a more stable physical state include the phase relationship between HR and blood pressure, as well as the baroreflex, which is a blood pressure sensor and modulator (Lehrer

Fig. 2 Example of a typical visual interface employed during HRV biofeedback training. The top panel shows sinusoidal waves associated with HR (red trace) and breathing (blue trace), demonstrating the systems' approach to coherence as breathing rate is actively altered. The power spectrum displayed confirms that HR frequency is peaking at 0.1 Hz, the reportedly optimal breathing frequency (re-printed with permission from Lehrer et al. 2014)



2013b). Taken together, evidence suggests that a breathing frequency of 0.1, or a breathing rate of 6 breaths per minute, typically results in maximum benefit (Vaschillo et al. 2006).

Achieving coherence using HRV biofeedback techniques requires the appropriate equipment, software, and training, but this can typically be obtained within a few minutes for naïve users (Lehrer et al. 2014; Vaschillo et al. 2006). Apart from HRV biofeedback training, other biofeedback approaches can incorporate minimal or no training, depending on how intuitive the format of feedback is. Using biofeedback and self-tracking of physiology to manage symptoms of acute and chronic stress to improve health and performance is becoming ubiquitous due to the affordability, sensitivity, and functionality of such devices in the form of high-grade consumer products. In high-risk settings, this type of passive monitoring allows for continuous data collection, without requiring a practitioner to stop a critical task, making it a highly valuable potential mechanism for performance feedback (Kennedy et al. under review).

The effectiveness of biofeedback has been documented in addressing psychiatric (Schoenberg et al. 2014) and physical (Khazan 2013) disorders, improving unhealthy symptoms and performance (Gevirtz 2013). This can partially be explained by operant conditioning mechanisms occurring during breathing-based biofeedback interventions such as HRV biofeedback. Notably, evidence has shown that volitional breathing behaviors can be modified via operant conditioning procedures such as biofeedback (Ley 1999). However, the effectiveness of biofeedback in enhancing health by managing one's stress levels, especially among non-patient populations, has received less attention, but is promising in concept (Lehrer 2007; Lehrer et al. 2008). There are a number of professionals in high-risk occupations, whose performance depends on appropriate management of stress under pressure, whose inappropriate management of stress can lead to dire consequences. These professionals, including physicians, police officers, and soldiers, represent individuals who stand to benefit the most from proper stress management (Frazier et al. 2018). Given that these individuals more frequently experience high-level acute stressors, we would expect to see the greatest benefit as a result of a stress reduction intervention among these populations.

To better understand the benefits of stress reduction techniques in high-risk professions, it is imperative to identify appropriate measures, and particularly the consistency of these measures across studies. Stress reduction techniques such as progressive muscle relaxation, different forms of meditation, cognitive-behavioral therapy, hypnosis, guided imagery, and biofeedback have been evaluated in work settings (Bormann et al. 2006), biofeedback has the advantage of providing objective guidance in a passive manner, thus incurring minimal interruptions into daily workflow. The effectiveness of stress reduction techniques such as

biofeedback can readily be evaluated through a variety of modalities, but if physiological, psychological, and/or performance indices and assessments are not consistent across studies, the ultimate implication is a lack of comparability. At the same time, standardizing performance assessment across all populations could contribute to a lack of sensitivity to specific contexts and task requirements expected of those populations. Using lab-based assessments primarily relied upon when recruiting convenience samples, for example, reduces the ecological validity when applied to deliberately recruited populations.

The goal of this systematic literature review is to evaluate the evidence of how biofeedback is currently being used to address stress management for the purposes of enhancing performance. Specifically, we are asking the following questions:

1. How does biofeedback affect physiological, psychological, and performance measures with respect to the type of population recruited? Which populations benefit the most from biofeedback interventions?
2. How consistent or standardized are the approaches used to answer these questions (including physiological indices and equipment, psychological constructs and instruments, and performance tasks and assessments)?
3. How do physiological, psychological, and performance outcomes change with respect to the frequency of the intervention and the duration of the study?

Our scope includes all populations of healthy individuals across any profession, with a special interest in high-risk professions. In the following description, we synthesize the literature to identify patterns and determine an effective approach based on the evidence available.

2 Methods

2.1 Search strategy

Articles were originally selected by searching EBSCOhost, JSTOR, PsycINFO, PubMed, and others using the university library search features. Search terms included [real-time AND biofeedback AND stress AND (professional OR practitioner)], and articles were included between January 1, 2000 and June 1, 2017. One researcher (LK) completed this step, ensuring that search terms scanned all available text, rather than searching within titles or keywords only. Additional filters were imposed to narrow the search to articles written in English and excluded magazine, trade publication, and newspaper articles. Following this compilation, duplicates acquired through more than one source were removed by one researcher (LK).

2.2 Inclusion/exclusion criteria and quality assessment

Abstracts were then reviewed according to pre-defined inclusion and exclusion criteria by one researcher (LK). Inclusion criteria ensured that resulting articles were limited to original research articles and included biofeedback administered during the task, used as the primary experimental procedure, and for stress management purposes explicitly. Exclusion criteria eliminated articles consisting of non-human and non-trial articles, as well as articles evaluating patient populations. Types of articles that were removed in this step included methods papers, reviews, instrument design articles, protocols, animal studies, machine uses, molecular approaches, case studies, and abstracts from conference proceedings.

All remaining articles underwent a quality assessment by one researcher (LK) to ensure rigorous methods and transparent reporting. Final articles were considered high quality and included in the review only if they described the biofeedback modality, participant/sample demographics, and intervention design and conditions. Final articles also had to report two of the following three data types: physiological, psychological, and performance measures. The final articles included in the review were approved independently by two researchers (LK, SHP) to ensure agreement (Fig. 3).

3 Results

Of the final 22 articles, the three articles written by Prinsloo and colleagues (Prinsloo et al. 2011, 2013a, b) report different sets of results from the same study. For the purposes of accurate evaluation, the results of these three articles are considered to represent one study (N of articles = 22; N of studies = 20).

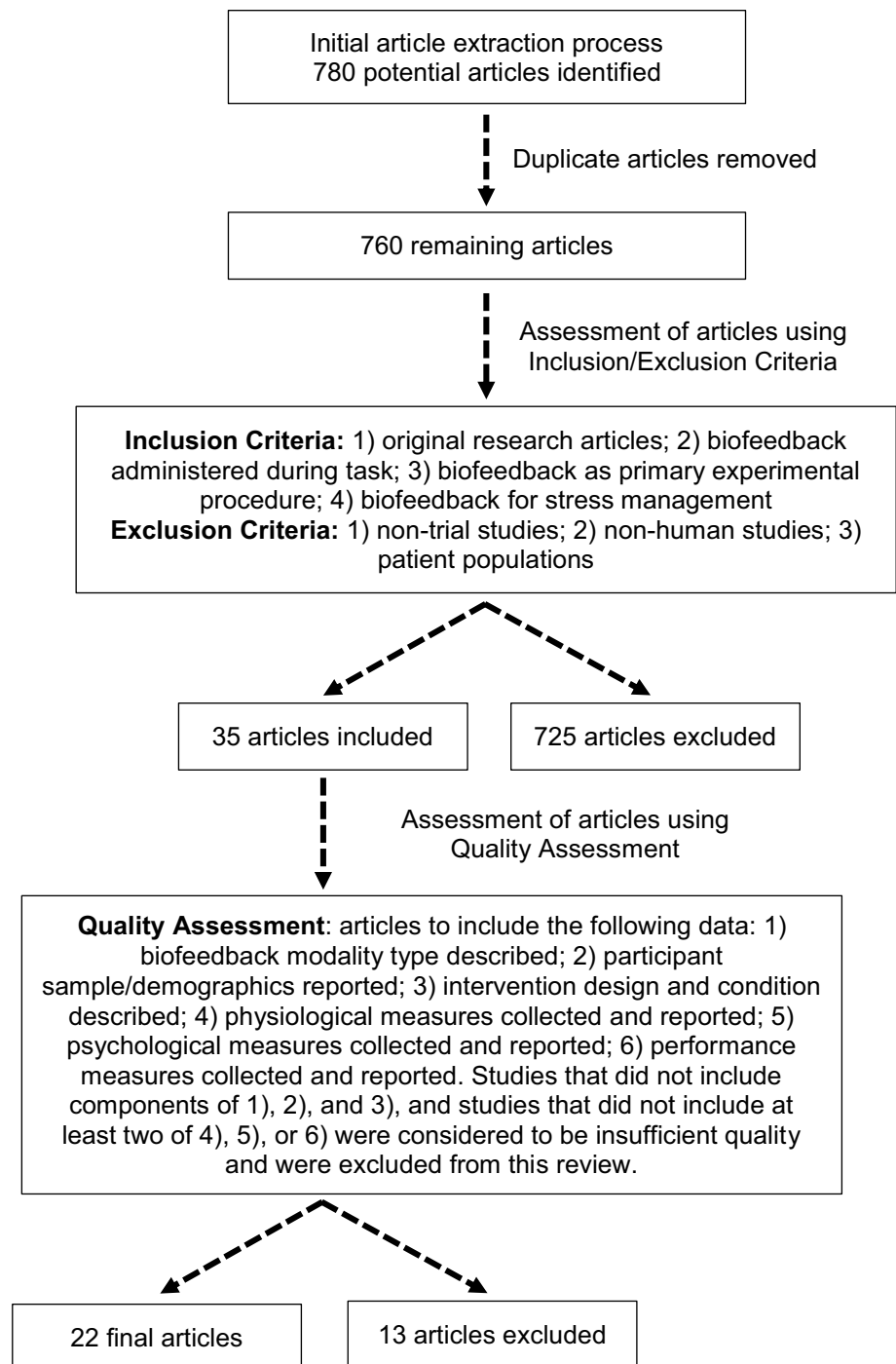
The 22 articles (20 studies) remaining almost exclusively evaluate and address mental, cognitive, or emotional stress. There was one exception, which evaluated and addressed the management of physical stressors among college students (Evetovich et al. 2007). Since this is the only article of its kind dealing with physical stress, we will report the results separately to exclusively capture mental stress in the following results. Evetovich and colleagues (2007) recruited 12 college students to evaluate the effectiveness of visual mechanomyographic biofeedback on forearm muscle flexion and relaxation. Performance was measured by calculating the number of repetitions of the seated preacher curl exercise, while physiological signals were measured using BioPac acquisition hardware and software to capture mechanomyographic and electromyographic amplitudes. This study took place over 9 days, and each participant completed 5 sessions total. Performance analysis revealed no significant

differences between the experimental and control groups, while physiological analysis revealed a decrease in both mechanomyographic and electromyographic amplitudes in the experimental vs. control groups.

Two of the 22 articles (20 studies) reported correlational results only, eliminating the ability to detect and report causal relationships between the biofeedback intervention and outcome measures (van Dijk et al. 2015; Zauszniewski et al. 2013). Because our focus is on hypothesis-driven, cause-and-effect results of introducing biofeedback, we will report these results separately. Van Dijk and colleagues (2015) recruited 74 healthy adults as participants to determine the synchronicity between physiological indicators of stress and self-perceived stress during and following a biofeedback intervention. Participants in the experimental groups completed a variety of short, computer-based, stress-inducing tasks while viewing their trending HR from the preceding 3 min, along with their HR value displayed in beats per minute (BPM). One experimental group was told that the display represented their current HR, while the other experimental group was informed that the display represented their current stress level. The control group completed the same tasks but did not receive physiological feedback. Analysis of physiological data revealed that for the group that was viewing their “current HR” in real time, there was a higher correlation between actual HR and their self-reported momentary and retrospective stress compared to the control group. Results of psychological assessments revealed that there was a positive correlation between the personality factor of neuroticism and momentary and retrospective stress in the control group. Furthermore, there was a positive correlation between the personality factor of anxiety sensitivity and momentary and retrospective stress in the experimental group viewing their “current stress” in real time. Their results broadly support the perspective that monitoring physiological signals may enhance body awareness, and that distinct personality factors may distinguish those individuals who benefit from biofeedback and those who may not.

Zauszniewski and colleagues (2013) recruited 20 grandmothers raising their grandchildren in an effort to reveal the relationships between physiological indicators of stress and self-perceived stress as a result of a biofeedback intervention. Participants used the handheld StressEraser to view their HRV waveform in real time with the goal of achieving coherence. A higher coherence index is known to be representative of a lower state of stress (Childre et al. 2010). Overall, analysis revealed a negative correlation between perceived stress and coherence, negative emotions and coherence, and depressive cognitions and coherence. These correlations indicate that as coherence decreases—a physiological indication of increasing stress—perceived stress, negative emotions, and depressive cognitions all increase.

Fig. 3 Search and elimination process



While this study does not purport or support the role of biofeedback directly leading to psychological changes, it does illustrate and validate the relationship between coherence and cognitive and emotional states.

By omitting the three studies reported separately, and combining the three Prinsloo articles into one study, the total number of studies evaluated in the following sections is 17 (see results below for detailed descriptions of full results,

separated into various categories of interest, and refer to Appendix A for comprehensive results of all studies)

3.1 Population studied

Of the 17 studies reviewed, there was a variety of populations recruited. For the purposes of our evaluation and discussion, we have separated the types of populations into convenience samples ($N=9$; 53%) and deliberately selected

samples ($N=8$; 47%). Convenience samples included high school students (Bradley et al. 2010; Kim et al. 2012), college students (Astor et al. 2013; Escolano et al. 2014; Henriques et al. 2011; Raaijmakers et al. 2013; Whited et al. 2014), and healthy adults (Cohen et al. 2016; Sherlin et al. 2010). Deliberate samples included correctional officers (McCraty et al. 2009), senior managers (Prinsloo et al. 2011, 2013a, b), soldiers (Bouchard et al. 2012), professional male baseball players (Sherlin et al. 2013), post-partum mothers (Kudo et al. 2014), police officers (Andersen et al. 2016), young male athletes (Dziembowska et al. 2016), and elite professional male soccer players (Rusciano et al. 2017). Individuals comprising samples in this latter category represent those whose optimal performance under acute stress is critical to execute their job successfully (Table 1).

In the following results, participant groups referred to as “experimental” include participants who received a biofeedback intervention, while groups referred to as “control” received either sham feedback, or no feedback at all. One study had an experimental group only (Sherlin et al. 2013), one study had a within-subjects design, where each participant served as their own control (Cohen et al. 2016), and another study used an immediate and delayed intervention design (Henriques et al. 2011). In this design, each group

served as a control group at one timepoint and an experimental group at the other timepoint, experiencing the same intervention at different times.

3.2 Types of biofeedback used

Across all studies included, 100% used the visual modality to convey biofeedback information, and 12% ($N=2$) additionally used the auditory modality to reinforce biofeedback (Bouchard et al. 2012; Sherlin et al. 2013). Both studies using visual and auditory biofeedback recruited specialized populations and delivered biofeedback in a way that abstractly represented physiological processes.

Since all studies, including these two, used the visual modality to represent the user’s physiology, we will further explore how this information was displayed. Overall, there were four total studies (24%) that described an abstract visual representation of underlying physiological state and changes in state (Astor et al. 2013; Bouchard et al. 2012; Escolano et al. 2014; Sherlin et al. 2013).

The most pervasive way of representing participants’ physiological state was by displaying direct physiological signals through the visual modality. This was done in 12 of the 17 studies (71%), and in each case, this signal was

Table 1 Study samples

References	Study population	Category of population	Sample (Expr/Ctrl)	Mean age (StDev)
McCraty et al. (2009)	Correctional officers	Deliberate	75 (43/32)	Expr: 39.47 (7.70) Ctrl: 40.72 (8.12)
Bradley et al. (2010)	High school students	Convenience	136 (77/59)	Expr: 15.3 (0.44) Ctrl: 15.3 (0.44)
Sherlin et al. (2010)	Healthy adults	Convenience	43	33.2 (8.77)
Henriques et al. (2011)	College students	Convenience	35	Not reported
Prinsloo et al. (2011, 2013a, b)	Senior managers	Deliberate	18 (9/9)	Expr: 33 (6) Ctrl: 34 (6)
Bouchard et al. (2012)	Soldiers	Deliberate	41 (21/20)	24.9 (5.55)
Kim et al. (2012)	High school students	Convenience	46 (21/25)	Not reported
Astor et al. (2013)	College students	Convenience	Study 1: 36 (19/17); Study 2: 68 (44/24)	Study 1: 23.39 (range: 8); Study 2: 22.06 (range: 9)
Sherlin et al. (2013)	Professional male baseball players	Deliberate	5	19.2 (0.45)
Raaijmakers et al. (2013)	College students	Convenience	28 (16/12)	22 (range: 8)
Escolano et al. (2014)	College students	Convenience	19 (10/9)	Expr: 25.8 (4.1) Ctrl: 24.3 (3.7)
Kudo et al. (2014)	Post-partum mothers	Deliberate	55 (25/30)	Expr: 30.5 (5.7) Ctrl: 33.4 (6.6)
Whited et al. (2014)	Undergraduate and graduate students	Convenience	28 (15/13)	Expr: 22.29 (3.6) Ctrl: 23.15 (4.06)
Andersen et al. (2016)	Police officers	Deliberate	12 (6/6)	31.5 (range: 7)
Cohen et al. (2016)	Healthy adults	Convenience	29	25.5 (4.67)
Dziembowska et al. (2016)	Young male athletes	Deliberate	41 (20/21)	18.34 (1.36)
Rusciano et al. (2017)	Professional male soccer players	Deliberate	20 (10/10)	Expr: 30.0 (3.8) Ctrl: 30.7 (4.3)

Provides details on the study sample and population

at least either HR, HRV, or HRV waveform (Bradley et al. 2010; Cohen et al. 2016; Dziembowska et al. 2016; Henriques et al. 2011; Kim et al. 2012; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Raaijmakers et al. 2013; Rusciano et al. 2017; Sherlin et al. 2010; Whited et al. 2014). Three of these 12 studies (25%; 18% of total 17 studies) presented the direct physiological signal, either as HR/HRV alone or in combination with other visual information. Of the 17 total studies, Cohen and colleagues were the only researchers to display direct indications of performance or error on the task (Cohen et al. 2016).

The remaining nine studies that presented physiology directly (75%; 53% of total 17 studies) displayed the HRV waveform and HRV biofeedback training geared towards achieving coherence (Bradley et al. 2010; Dziembowska et al. 2016; Henriques et al. 2011; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2010; Whited et al. 2014). Whited and colleagues (2014) additionally displayed HR in BPM and the HRV power spectrum to participants, while Rusciano and colleagues (2017) included information regarding skin conductance level (SCL), electromyography signal, and skin temperature signal in addition to the HRV waveform and cognitive exercises. The remaining study provided no details on the visual display (Andersen et al. 2016; Table 2).

3.3 Physiological equipment

Of the nine studies presenting the participants' HRV waveform back during task completion, five of those (56%; 29% of total 17 studies) used Freeze-Framer/emWave to display physiological information (Bradley et al. 2010; Dziembowska et al. 2016; Henriques et al. 2011; McCraty et al. 2009; Whited et al. 2014). Three of these studies (33%; 18% of total 17 studies) used StressEraser to display information back to the participant (Kudo et al. 2014; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010). Most of these studies used the devices listed to collect, process, and display information in real time, but Prinsloo and colleagues (2011, 2013a, b) collected and processed data using BioPac equipment and fed that data back using StressEraser, while Whited and colleagues (2014) collected and processed data using Polar RS800CS equipment and fed that data back using emWave. The remaining study that presented HRV waveform as the visual indicator of physiology used the Nexus 10 Mark II system to collect, process, and present physiological data (Rusciano et al. 2017).

Five of the total 17 studies (29%) used chest-worn HR belts to collect HR data (Andersen et al. 2016; Astor et al. 2013; Bouchard et al. 2012; Cohen et al. 2016; Whited et al. 2014). Zephyr (Andersen et al. 2016; Cohen et al. 2016) and Polar (Bouchard et al. 2012; Whited et al. 2014) were

equally popular among these five studies, with one remaining unspecified (Astor et al. 2013; Table 3).

3.4 Physiological measures and results

Across the 17 studies reviewed, there was a wide variety of physiological measures collected and analyzed by researchers (Table 4). The most common were HR/HRV, measured in 14 of the 17 total studies (82%), and electroencephalography (EEG), measured in 6 of the 17 (35%).

Of the 14 HR/HRV studies, 11 (79%) reported significant differences in the experimental group in at least one cardiovascular measure (Andersen et al. 2016; Astor et al. 2013; Bouchard et al. 2012; Bradley et al. 2010; Dziembowska et al. 2016; Kim et al. 2012; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Whited et al. 2014). Four of these 11 studies reported a main effect of time, such that the experimental group saw significant changes at the post-intervention timepoint compared to the pre-intervention timepoint (Dziembowska et al. 2016; Kim et al. 2012; McCraty et al. 2009; Rusciano et al. 2017). Four studies also reported a main effect of group, such that the experimental group saw significantly different changes at the post-intervention timepoint (Andersen and Gustafsson 2016; Rusciano et al. 2017) or during the intervention (Astor et al. 2013; Prinsloo et al. 2011, 2013b). The remaining four studies reported interaction effects, such that HR/HRV values saw significant changes among the experimental group compared to the control group at the post-intervention timepoint compared to a baseline or pre-intervention timepoint (Bouchard et al. 2012; Bradley et al. 2010; Kudo et al. 2014; Whited et al. 2014). Two studies (14%) reported no significant differences in HR/HRV (Henriques et al. 2011; Raaijmakers et al. 2013), and one study (7%) did not report the data (Cohen et al. 2016).

Other HR/HRV indices reported with significant changes include: arousal level (Astor et al. 2013); R-R intervals (Kim et al. 2012); low frequency/high frequency ratio (McCraty et al. 2009); root mean square of the successive differences (RMSSD; Prinsloo et al. 2011, 2013a, b); percentage of normal-to-normal peaks with a difference greater than 50 ms (pNN50; Whited et al. 2014); coherence index (Bradley et al. 2010; Dziembowska et al. 2016); standard deviation from normal-to-normal (SDNN; Bradley et al. 2010; Kudo et al. 2014; Prinsloo et al. 2011, 2013a, b); and low frequency power (Bradley et al. 2010; Dziembowska et al. 2016; Rusciano et al. 2017), see Table 3 for a detailed account of physiological measures and results.

Of the six studies measuring EEG, whether using quantitative EEG (Sherlin et al. 2010, 2013), traditional EEG (Escalano et al. 2014; Prinsloo et al. 2011, 2013a, b), or alpha asymmetry (Dziembowska et al. 2016; Raaijmakers et al. 2013), five saw significant differences (83%) within at

Table 2 Biofeedback modality

Reference	Modality: v:visual	Category of visual display	Modality:auditory
McCraty et al. (2009)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Bradley et al. (2010)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Sherlin et al. (2010)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Henriques et al. (2011)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Prinsloo et al. (2011, 2013a, b)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Bouchard et al. (2012)	Red texture partially obscuring user's field of view. Increased arousal represented by an increasingly obstructed view	Abstract representation(s)	Sound of a pumping heart increasing in frequency and loudness with increased arousal
Kim et al. (2012)	ECG tachogram, anxiety profile	Direct physiological signal(s)	N/A
Astor et al. (2013)	Color-coded arousal meter; corresponding game elements changing color to reflect the state of arousal, located directly in attentional field	Abstract representation(s)	N/A
Sherlin et al. (2013)	Dual Drive Xtreme: car presented in a race, moves at a speed proportional to user success; or Particle Editor: shapes or lines in movement along with music when parameters are met	Abstract representation(s)	Music
Raaijmakers et al. (2013)	Across 6 tasks: 3 displayed SCL signal, 2 displayed HRV signal, 1 displayed a combination of HRV and SCL signals	Direct physiological signal(s)	N/A
Escolano et al. (2014)	A square with changing saturation colors; physiological values reflecting baseline appear gray, values above baseline appear red and increase in saturation as values increase, values below baseline appear blue and increase in saturation as values decrease	Abstract representation(s)	N/A
Kudo et al. (2014)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Whited et al. (2014)	HR, HRV power spectrum, HRV waveform (coherence)	Direct physiological signal(s)	N/A
Andersen et al. (2016)	No details	No details	N/A
Cohen et al. (2016)	Performance prediction bar graph, HR bar graph, error chance prediction bar graphs	Direct physiological signal(s)	N/A
Dziembowska et al. (2016)	HRV waveform (coherence)	Direct physiological signal(s)	N/A
Rusciano et al. (2017)	Sessions 1–3: HRV waveform (coherence); Sessions 4–9: HRV waveform (coherence), SCL level, EMG signal, skin temperature signal; Sessions 10–15: HRV waveform (coherence)	Direct physiological signal(s)	N/A

Provides details on the presentation structure of visual and auditory forms of biofeedback

HRV heart rate variability, *ECG* electrocardiography, *SCL* skin conductance level, *EMG* electromyography

least one component in the experimental group compared to the control group. Raaijmakers and colleagues were the only group reporting no significant differences (Raaijmakers et al. 2013). Each of the five studies reporting significant differences between experimental and control groups reported changes in distinct bandwidths and neural areas.

Of the measures collected and evaluated in only one study, many of those showed statistically significant improvements

when comparing levels before the intervention to afterwards, in the experimental group alone. These measures included dehydroepiandrosterone, secretory immunoglobulin A, cholesterol, triglycerides, fasting glucose levels (McCraty et al. 2009), respiratory frequency (Prinsloo et al. 2011, 2013a, b), and surface electromyography, skin temperature, and blood volume pulse (BVP; Rusciano et al. 2017).

Table 3 Physiological measures, equipment, and results

References	Physio measures	Physio equipment	Physio results
McCraty et al. (2009)	Cortisol, DHEA, S-IgA cholesterol, triglycerides, fasting glucose levels, HR, HRV, BP	Freeze-Framer/emWave	Pre-post Expr: ↓ total cholesterol, LDL, total/HDL ratio, glucose levels, DHEA, mean arterial pressure, systolic BP, diastolic BP, mean HR, ↑ LF/HF ratio of HRV; Pre-post Ctrl: ↓ total cholesterol, HDL, glucose levels
Bradley et al. (2010)	HRV	Freeze-Framer/emWave	Pre-post Expr vs. Ctrl: ↓ HR, ↑ SDNN, HF, LF, IC
Sherlin et al. (2010)	QEEG	StressEraser	Pre-post Expr: ↑ overall alpha; Pre-post Ctrl: ↓ overall alpha
Henriques et al. (2011)	HRV	emWave	Pre-post Expr: no significant differences
Prinsloo et al. (2011, 2013a, b)	BP, EEG, HR, HRV, RR, RF	BioPac, StressEraser	Interv. Expr vs. Ctrl: ↓ RR, RF, ↑ LF power, TF power, (2011), ↓ relative frontal and central beta, ↑ SDNN, relative frontal theta, frontal, central, and parietal theta/beta ratio (2013b) Pre-post Expr: ↓ HR (2011); Pre-post Ctrl: ↓ HR (2011)
Bouchard et al. (2012)	Daily salivary cortisol, HR	Salivette, Polar wireless transmitter belt	Post Expr vs. Ctrl: ↓ cortisol; Mid Expr vs. Ctrl: ↓ HR
Kim et al. (2012)	HRV	QECG-3	Pre-post Expr: ↑ R-R intervals
Astor et al. (2013)	HR (arousal level)	Chest-worn HR belt	Study 1 Expr vs. Ctrl: not reported; Study 2 Expr vs. Ctrl: ↓ arousal level
Sherlin et al. (2013)	QEEG	Electro-Cap International	Pre-post: ↑ central beta (4/5), frontal beta (3/5), parietal beta (3/5)
Raaijmakers et al. (2013)	HRV, SCL, EEG alpha asymmetry	ActiveTwo	Pre-post Expr vs. Ctrl: No significant differences
Escolano et al. (2014)	EEG	Guger Technologies	Pre-post Expr: ↑ UA power, LA2
Kudo et al. (2014)	HRV	StressEraser	Pre-post Expr vs. Ctrl: ↓ HR, ↑ SDNN, HF power
Whited et al. (2014)	HR, HRV	Polar RS800CS, emWave	Pre-post Expr vs. Ctrl: ↑ pNN50 stress reactivity
Andersen et al. (2016)	HR, BP	Zephyr BioHarness, Omnisense ambulatory blood pressure wrist cuffs	Post Expr vs. Ctrl: ↓ HR max between baseline and Scenario 1, ↓ change in HR from baseline to Scenario 1
Cohen et al. (2016)	HR, HRV	Zephyr HxM	Not reported
Dziembowska et al. (2016)	EEG alpha asymmetry, HRV	emWave, Electro-Cap International	Pre-post Expr: ↑ HRV spectral power, LF HRV, IC, theta and alpha spectral power, alpha asymmetry, ↓ HF HRV
Rusciano et al. (2017)	SCL, RR, BVP, HR, HRV, surface EMG, skin temperature	Nexus 10 Mark II	Post Expr vs. Ctrl: ↓ SCL, HR, RR; Pre-post Expr: ↓ RR ↑ LF

Provides details regarding physiological equipment, measures, and results

In the Results column, “Pre-post Expr” refers to a main effect of time, where the experimental group experienced the associated significant changes in physiological measures (in the direction indicated by the arrow) after receiving a biofeedback intervention compared to beforehand. “Pre-post Ctrl” indicates that significant differences were observed in the control group at the second timepoint compared to the first timepoint, despite the lack of a biofeedback intervention in between. “Post Expr vs. Ctrl” indicates a main effect of group was reported, such that the experimental group saw significant changes compared to the control group at the post-intervention timepoint. “Pre-post Expr vs. Ctrl” refers to the presence of an interaction effect, where the experimental group experienced significantly different results after the intervention compared to the control group. “Interv Expr vs. Ctrl,” appearing only once (Prinsloo et al. 2011, 2013b), refers to a main effect of group, where the physiological measures acquired during the intervention were significantly different for the experimental group compared to the control group. One study (Bouchard et al. 2012) reported a main effect of group such that the experimental group had significantly different values compared to the control group at a mid-point in the design (referred to as “Mid Expr vs. Ctrl”). “Pre-post,” also appearing only once (Sherlin et al. 2013), refers to the lack of a control group

DHEA dehydroepiandrosterone, *S-IgA* secretory immunoglobulin, *HR* heart rate, *HRV* heart rate variability, *BP* blood pressure, *LDL* low-density lipoprotein cholesterol, *HDL* high-density lipoprotein cholesterol, *LF* low frequency power, *HF* high frequency power, *SDNN* standard deviation from normal-to-normal, *IC* coherence index, *QEEG* quantitative electroencephalography, *EEG* electroencephalography, *RR* respiratory rate, *RF* respiratory frequency, *TF* total frequency power, *RMSSD* root mean square of the successive differences, *QECG-3* quantitative electrocardiography—three limb lead, *SCL* skin conductance level, *UA* upper alpha bandwidth, *LA2* lower alpha 2 bandwidth, *pNN50* percentage of normal-to-normal peaks with a difference greater than 50 ms, *BVP* blood volume pulse, *EMG* electromyography

Table 4 Description of components within physiological measures and psychological assessments across studies with significant changes reported

Variables	Meaning
Physiological measures	
Heart rate	Average heartbeats per minute over a time window
R-R intervals	Variability among R–R intervals from ECG data provide heart rate
Low frequency power	HRV power spectrum ranging from 0.04 to 0.15 Hz
High frequency power	HRV power spectrum ranging from 0.15 to 0.4 Hz
Low frequency/high frequency ratio	Ratio of low frequency power to high frequency power
Total frequency power	HRV power spectrum ranging from 0.005 to 0.4 Hz
Root mean square of the successive differences	HRV time-domain measure calculating the variation among successive R–R intervals
Standard deviation from normal-to-normal	HRV time-domain measure calculating the variation of R-R interval values in reference to a mean
pNN50	HRV time-domain measure calculating the percentage of successive R-R intervals that differ by 50 ms or more
Coherence index	Primary marker of psychophysiological coherence state, derived from the degree of symmetry of the HRV waveform; a higher value indicates higher symmetry and a more positive state
Delta bandwidth	EEG power spectrum ranging from 1 to 3.5 Hz
Theta bandwidth	EEG power spectrum ranging from 4 to 7.5 Hz
Alpha bandwidth	EEG power spectrum ranging from 8 to 12 Hz
Beta bandwidth	EEG power spectrum ranging from 13 to 32 Hz
Gamma bandwidth	EEG power spectrum ranging from 32 to 45 Hz
Theta/Beta ratio	Ratio of theta spectral power to beta spectral power
Skin conductance level	Measure of the degree of electrical conduction across the skin, determined by activity of eccrine sweat glands
Respiratory rate	Average breaths per minute over a time window
Cortisol	Stress-related hormone, measurable by saliva or blood sample collection
Low-density lipoprotein (LDL) cholesterol	Transporter of fat molecules; contributes to fatty build-up in arteries when elevated
High-density lipoprotein (HDL) cholesterol	Transporter of LDL cholesterol from arteries to the liver, where it is broken down
Triglycerides	Most common type of fat in the body; when measured in blood, high levels are an indicator of poor health
Total cholesterol	Measure of overall cholesterol levels, derived by summing LDL, HDL and triglyceride levels
Blood glucose levels	Overall level of sugar in the blood; high levels are an indicator of poor health
Dehydroepiandrosterone	Stress-related hormone
Secretory immunoglobulin A	Mucosal class of antibody with protective immunological qualities
Mean arterial pressure	The average blood pressure within an individual during one cardiac cycle
Systolic blood pressure	Total pressure in blood vessels while the heart beats
Diastolic blood pressure	Total pressure in blood vessels between heartbeats
Psychological measures	
Brief Symptom Inventory	53-item questionnaire measuring psychological symptom patterns with 9 symptom scales: somatization, obsessive–compulsive, interpersonal sensitivity, depression, anxiety, hostility, phobic anxiety, paranoid ideation, and psychoticism; and 3 global indices: global severity index, positive symptom distress index, and positive symptom total
State-trait anxiety inventory for adults	20-item questionnaire evaluating the stable trait of stress; optional 20-item questionnaire evaluating the momentary state of stress
Personal wellness profile	75-item questionnaire assessing health information, physical activity, eating practices, alcohol, drugs, and smoking, stress and coping, social health, safety, medical care, and health view
Jenkins activity survey	Measure of Type A and coronary-prone behavior with three subscales: speed and impatience, job involvement, and hard-driving and competitive
Personal and organizational quality assessment	Measure of psychological and workplace contributors to organizational climate with 2 main areas: personal scales and organizational scales
Test anxiety inventory	16-item questionnaire measuring test anxiety globally, with 2 additional measurements of worry and emotionality
Perceived stress scale	10-item questionnaire assessing the global measure of self-perceived stress

Table 4 (continued)

Variables	Meaning
Mood and anxiety symptom questionnaire	90-item questionnaire measuring mood and anxiety symptoms with 6 subscales: general distress mixed, general distress anxious, general distress depressed, anxious arousal, loss of interest, and high positive affect
Scales of psychological well-being	54-item questionnaire measuring aspects of psychological functioning with 6 subscales: self-acceptance, positive relations with others, autonomy, environmental mastery, purpose in life, and personal growth
Smith relaxation states inventory 3	38-item questionnaire assessing relaxation states with 19 total relaxation states falling into 4 categories: mindfulness, energized positive feelings, basic relaxation, and transcendence
Perceived self-efficacy scale*	4-item questionnaire measuring confidence in oneself to control stress at different intensities
Simulator sickness questionnaire	16-item questionnaire measuring side effects induced by immersive simulation
Appreciation and usefulness of program*	13-item questionnaire evaluating perceptions of the intervention
Emotion regulation questionnaire	10-item questionnaire measuring emotion regulation, with 2 subscales: reappraisal and suppression
NeuroPerformance profile	Measures contributors to mental/cognitive aspects of athletic performance, with 3 indices: focus index, stress index, and speed index
Positive and negative affect scale	20-item questionnaire evaluating affect, with 2 subscales: positive affect and negative affect
Edinburgh postnatal depression score	10-item questionnaire measuring probable post-partum depression in mothers 8-week post-childbirth
Multiple adjective affect checklist-revised	66-item instrument assessing state affect, with 5 subscales: anxiety, depression, hostility, positive affect, and sensation seeking
Stress, skills, ability to manage scenarios^	Measures the anticipation of stress, self-confidence in decision-making, self-confidence in choosing the correct actions
Situational awareness^	Measures the degree of and confidence in perceived situational awareness
Level-of-Information processing scale	Measures the self-perceived task demand of the situation
System usability scale	10-item questionnaire measuring perceived usability of the feedback types
Rosenberg self-esteem scale	10-item questionnaire evaluating self-esteem or positive and negative feelings about the self

Provides a description of each physiological measure collected and psychological evaluation given across studies

*Questionnaires were developed by researchers specifically for their study and their demographic (Bouchard et al. 2012)

^Questions were developed by researchers specifically for their study and their demographic (Andersen et al. 2016)

Additionally, cortisol (Bouchard et al. 2012; McCraty et al. 2009), respiratory rate (RR; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017), and SCL (Raaijmakers et al. 2013; Rusciano et al. 2017) were evaluated in two studies each (12%), while blood pressure (BP) was evaluated in three studies (18%; Andersen et al. 2016; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b). Cortisol levels were significantly reduced in the experimental group compared to the control group in one of the two studies in which it was evaluated (Bouchard et al. 2012). RR was also significantly reduced in the experimental group compared to the control group, in both studies (Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017). Analysis of SCL showed a significant reduction in the experimental group when comparing levels before and after the intervention, reported in one of the two studies evaluating SCL (Rusciano et al. 2017). BP decreased significantly within the experimental group compared to the control group in one of the three studies (McCraty et al. 2009a, b).

3.5 Psychological instruments, measures, and results

The majority of studies reviewed, 14 of 17 (82%), included psychological assessments via self-report instruments. While there was some consistency in the types of measures collected, there was little overlap among the instruments used. The most frequently evaluated psychological measures included clinical symptoms, sleep, and stress/anxiety (Table 5, Appendix A).

Clinical symptoms (McCraty et al. 2009a, b; Whited et al. 2014) and sleep (Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2013) were each assessed across two studies (12% of those evaluating psychological measures). Both studies evaluating clinical symptoms used the Brief Symptom Inventory (Derogatis et al. 2012), and only one of those saw significant improvement from pre- to post-intervention assessment in the experimental group (McCraty et al. 2009). Sleep was evaluated differently in each of the studies, where it was reported, but in both cases, there was improvement (Prinsloo

Table 5 Psychological measures, instruments, and results

References	Psych measures	Psych instruments	Psych results
McCraty et al. (2009)	Health risk, Type A and coronary-prone behavior, psychological symptoms, psychological and workplace elements	Personal wellness profile, jenkins activity survey, brief symptom inventory, personal and organizational quality assessment	Pre-post Expr: ↓ anger, fatigue, hostility, interpersonal sensitivity, paranoid ideation, psychological distress, Type A behavior, speed and impatience, ↑ productivity, motivation, goal clarity, perceived manager support, gratitude, positive outlook; Pre-post Ctrl: ↓ work attitude, confidence in organization, ↑ depression
Bradley et al. (2010)	Test anxiety (global, worry, emotionality)	Test anxiety inventory (8 of 16 items)	Pre-post Expr vs. Ctrl: ↓ overall test anxiety (global, worry, emotionality)
Sherlin et al. (2010)	State anxiety	State-Trait Anxiety Interview-State Form	Pre-post Expr vs. Ctrl: ↓ state anxiety
Henriques et al. (2011)	Mood/anxiety symptoms, state and trait anxiety, subjective well-being	Mood and anxiety symptom questionnaire, state-trait anxiety inventory, scales of psychological well-being	Pre-post Expr: ↓ general distress: mixed, general distress: anxious, anxious arousal
Prinsloo et al. (2011, 2013a, b)	State and trait anxiety, relaxation, sleepiness	State-Trait Anxiety Inventory Forms Y-1 and Y-2, Smith Relaxation States Inventory 3, Visual Analog Scale	Pre-post Expr vs. Ctrl: ↑ relaxation, ↓ sleepiness
Bouchard et al. (2012)	Perceived self-efficacy to control stress	Perceived self-efficacy, Simulator Sickness Questionnaire, appreciation and usefulness of program	Pre-post Expr vs. Ctrl: ↑ perceived self-efficacy to control stress
Astor et al. (2013)	Cognitive reappraisal and suppression	Emotion Regulation Questionnaire	Not reported
Sherlin et al. (2013)	Focus index, stress index, speed index	NeuroPerformance Profile	Pre-post: ↑ sleep, focus, feeling of being relaxed, all indices of NeuroPerformance Profile
Raaijmakers et al. (2013)	Affect	Positive and Negative Affect Scale	Pre-post Expr vs. Ctrl: no significant differences
Kudo et al. (2014)	Postnatal depression	Edinburgh Postnatal Depression Score (EPDS)	Pre-post Expr vs. Ctrl: ↓ EPDS score, anxiety, difficulty sleeping, sad and miserable thoughts
Whited et al. (2014)	Distress, stress, state affect	Brief Symptom Inventory, Perceived Stress Scale, Multiple Adjective Affect Checklist-Revised	Pre-post Expr vs. Ctrl: no significant differences
Andersen et al. (2016)	Perceived stress, skills, ability to manage scenarios, self-confidence in decision-making, situational awareness	Stress, Skills, Ability to manage scenarios, Situational awareness	Pre Ctrl vs. Expr: ↑ confidence
Cohen et al. (2016)	Appraisal, task demand, usability	Likert-type scale, Level-of-Information Processing Scale, System Usability Scale	HR vs. noHR: ↓ perceived usability of feedback
Dziembowska et al. (2016)	State anxiety, self esteem	State-Trait Anxiety Inventory A-State Scale, Rosenberg Self Esteem Scale	Pre-post Expr: ↓ state anxiety

Provides details regarding psychological instruments, measures, and results for the studies assessing psychological constructs. In the Results column, “Pre-post Expr” refers to a main effect of time, where the experimental group experienced the associated significant changes on self-reported measures (in the direction indicated by the arrow) after receiving a biofeedback intervention compared to beforehand. “Pre-post Ctrl” indicates that significant differences were observed in the control group at the second timepoint compared to the first timepoint, despite the lack of a biofeedback intervention in between. “Post Expr vs. Ctrl” indicates a main effect of group was reported, such that the experimental group saw significant changes compared to the control group at the post-intervention timepoint. One study reported a main effect of group reflecting an significantly higher value self-reported measure in the control group at the pre-intervention timepoint (noted as “Pre Ctrl vs. Expr”; Andersen et al. 2016). “Pre-post Expr vs. Ctrl” refers to the presence of an interaction effect, where the experimental group experienced significantly different results after the intervention vs. beforehand, compared to the control group. “HR vs. noHR,” appearing only once (Cohen et al. 2016), refers to a main effect of group, where the self-reported responses recorded, while heart rate feedback was available were significantly different for that same group of participants in a different condition, while heart rate feedback was not available (within-subjects experimental design). “Pre-post,” also appearing only once (Sherlin et al. 2013), refers to the lack of a control group

et al. 2011, 2013a, b; Sherlin et al. 2013). This improvement was demonstrated by an interaction effect in one study, whereby the experimental group reported less sleepiness after the intervention vs. before, compared to the control group (Prinsloo et al. 2011, 2013a, b), and as a main effect of time for the experimental group in one other study (Sherlin et al. 2013).

The most consistently evaluated measure was stress/anxiety, assessed in 8 of the 14 studies (57%) measuring psychological constructs (Andersen et al. 2016; Bradley et al. 2010; Dziembowska et al. 2016; Henriques et al. 2011; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010, 2013; Whited et al. 2014). Four of these eight studies used the State-Trait Anxiety Inventory for Adults (Spielberger et al. 1983) to assess the construct of stress (Dziembowska et al. 2016; Henriques et al. 2011; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010). The remaining four studies evaluating stress/anxiety used unique instruments. Of the eight studies evaluating stress/anxiety, five of these (62.5%) saw a significant decrease in stress and/or anxiety from pre- to post-intervention in the experimental group (Bradley et al. 2010; Dziembowska et al. 2016; Henriques et al. 2011; Sherlin et al. 2010, 2013).

Aside from the instruments mentioned, there were no additional standardized assessments appearing in more than one study. Overall, there were 23 unique scales or inventories used across the 14 studies that evaluated psychological constructs (Table 5).

3.6 Performance instruments, measures, and results

Twelve of the 17 total studies (71%) evaluated performance (Andersen et al. 2016; Astor et al. 2013; Bouchard et al. 2012; Bradley et al. 2010; Cohen et al. 2016; Escolano et al. 2014; Kim et al. 2012; Prinsloo et al. 2011, 2013a, b; Raaijmakers et al. 2013; Rusciano et al. 2017; Sherlin et al. 2010, 2013). Across these 12 studies, there were only two performance indices and one assessment that was repeated more than once.

While 8 of the 12 studies evaluating performance (67%) used at least reaction time (RT) and/or errors made to approximate task performance (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b; Raaijmakers et al. 2013; Rusciano et al. 2017; Sherlin et al. 2010, 2013), five total studies (42%) evaluated performance in a way that was more specialized to the populations recruited (Andersen et al. 2016; Bouchard et al. 2012; Bradley et al. 2010; Kim et al. 2012; Rusciano et al. 2017). Of those evaluating RT and errors made, four (50%) used a version of the Stroop task (Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2010). The other four studies evaluating performance via

RT and errors used unique tasks to measure those metrics (Astor et al. 2013; Cohen et al. 2016; Raaijmakers et al. 2013; Sherlin et al. 2013). Six out of these eight tasks (75%) saw improvements in some component as a result of the intervention (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2013).

Four of the 12 studies assessing performance (33%) included performance measures or tasks related to the deliberate population recruited (Andersen et al. 2016; Bouchard et al. 2012; Rusciano et al. 2017; Sherlin et al. 2013). Notably, these four studies comprise half of the total studies recruiting specialized or deliberate samples. All four of these studies reported improvements in some aspect of performance as a result of the intervention. In contrast, three of the 12 studies (25%) saw no improvements (Bradley et al. 2010; Raaijmakers et al. 2013; Sherlin et al. 2010), all of which recruited participants using convenience sampling methods. Overall, nine of the 12 studies evaluating performance (75%) saw some improvements (Andersen et al. 2016; Astor et al. 2013; Bouchard et al. 2012; Cohen et al. 2016; Escolano et al. 2014; Kim et al. 2012; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2013; Table 6).

3.7 Features of the intervention

Session duration had little variety across studies. Of the 17 studies, 13 total (76%) consisted of session durations that lasted for less than 1 h (Astor et al. 2013; Bouchard et al. 2012; Dziembowska et al. 2016; Escolano et al. 2014; Henriques et al. 2011; Kim et al. 2012; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Raaijmakers et al. 2013; Rusciano et al. 2017; Sherlin et al. 2010, 2013; Whited et al. 2014), with the longest duration of those being 35 min (Sherlin et al. 2013). Only one of the studies was longer than 1 h (6%), ranging from 90 to 120 min (Cohen et al. 2016). Additionally, one study (6%) reported the session duration as variable, depending on how long it took the participant to achieve a certain objective (Kudo et al. 2014). Two of the 17 studies (12%), however, provided no detail regarding session duration (Andersen et al. 2016; Bradley et al. 2010).

The frequency of sessions over the course of the study was more variable across studies. The experimental session for five of the 17 studies (29%) occurred only once (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010). Eleven of the 17 studies (65%) reported multiple sessions (Bouchard et al. 2012; Bradley et al. 2010; Dziembowska et al. 2016; Escolano et al. 2014; Henriques et al. 2011; Kim et al. 2012; Kudo et al. 2014; McCraty et al. 2009; Raaijmakers et al. 2013; Rusciano et al. 2017; Sherlin et al. 2013; Whited et al. 2014). Of those studies with repeated sessions, five of the eleven (45%) had between three and ten total sessions

Table 6 Performance measures, instruments, and results

References	Performance measures	Performance instruments	Performance results
Bradley et al. (2010)	Standardized testing scores	California high school exit exam, california standards test	Pre-post Expr vs. Ctrl: No significant differences
Sherlin et al. (2010)	Incorrect vs. correct response, RT, missed responses	Modified Stroop task	Pre-post Expr vs. Ctrl: No significant differences
Prinsloo et al. (2011, 2013a, b)	RT, mistakes made	Stroop task	Pre-post Expr vs. Ctrl: ↑ reaction time, consistency in responses, ↓ mistakes made
Bouchard et al. (2012)	Critical task completion, program elements met	Trainees Evaluation Sheet, ImpACT coach checklist	Post Expr vs. Ctrl: ↑ performance
Kim et al. (2012)	Speech performance	Korean Broadcasting System speech performance test	Pre-post Expr vs. Ctrl: ↑ speech performance
Astor et al. (2013)	Percentage of correct decisions	Auction Game decision performance	Study 1 Expr vs. Ctrl: ↑ performance; Study 2 Expr vs. Ctrl: ↑ performance
Sherlin et al. (2013)	RT, response time variability, errors of omission, errors of commission	QIKtest continuous performance test	Pre-post: ↓ errors of commission
Raaijmakers et al. (2013)	RT, error percentages	N-back task, Mental Rotation Task	Pre-post Expr: ↓ RT, ↑ accuracy Pre-post Ctrl: ↓ RT, ↑ accuracy
Escolano et al. (2014)	Errors, elapsed time, recognized words, interference, correct responses, RT	Paced Auditory Serial Addition Task (PASAT), Rey Auditory Verbal Learning Test (RAVLT), Trail Making Test (TMT), Stroop Color-Word Test, Mental Rotation Task (MRT)	Pre-post Expr vs. Ctrl: ↑ TMT part B; Pre-post Expr: ↓ errors and time elapsed in PASAT, RT in MRT, ↑ recognized words in RAVLT, correct responses in MRT; Pre-post Ctrl: ↓ time elapsed in PASAT, ↑ correct responses in MRT, TMT part A
Andersen et al. (2016)	Critical task completion, situational awareness, decision-making, overall performance	Senior trainers' ratings	Pre-post Expr vs. Ctrl: ↑ critical task completion; Post Expr vs. Ctrl: ↑ overall performance, situational awareness, decision-making
Cohen et al. (2016)	Total score for task, number of errors made	Simulated fire management task	Expr vs. Ctrl: ↑ relative performance score
Rusciano et al. (2017)	Accuracy, RT, injury prevention	Stroop task, visual search task	Pre-post Expr vs. Ctrl: ↑ accuracy in Stroop task, presence at training days ↓ RT in visual search task, absences at training days, injuries

Provides details regarding performance instruments, measures, and results for the studies assessing performance (RT: reaction time). In the Results column, "Pre-post Expr" refers to a main effect of time, where the experimental group experienced the associated significant changes in some aspect of performance (in the direction indicated by the arrow) after receiving a biofeedback intervention compared to beforehand. "Pre-post Ctrl" indicates that significant differences were observed in the control group at the second timepoint compared to the first timepoint, despite the lack of a biofeedback intervention in between. "Post Expr vs. Ctrl" indicates a main effect of group was reported, such that the experimental group saw significant changes compared to the control group at the post-intervention timepoint. "Pre-post Expr vs. Ctrl" refers to the presence of an interaction effect, where the experimental group experienced significantly different results after the intervention vs. beforehand, compared to the control group. One study used a within-subjects design with multiple experimental conditions and multiple control conditions per participant (Cohen et al. 2016). Results are presented as "Expr vs. Ctrl" to indicate all experimental conditions contrasted with all control conditions. "Pre-post," also appearing only once (Sherlin et al. 2013), refers to the lack of a control group

(Bouchard et al. 2012; Dziembowska et al. 2016; Kim et al. 2012; Raaijmakers et al. 2013; Whited et al. 2014), while six of the eleven (55%) held 15 or more sessions (Bradley et al. 2010; Henriques et al. 2011; Kudo et al. 2014; McCraty et al. 2009; Rusciano et al. 2017; Sherlin et al. 2013), with the greatest number of sessions approaching 40 (Bradley et al. 2010). One study (6%) had no details regarding the frequency of experimental sessions (Andersen et al. 2016).

Despite the relative consistency among session duration and frequency of intervention across studies, there was a much greater variety in terms of study duration. Study duration ranged from 1-day sessions to 5 months. Seven of the 17 total studies (41%) ran for a period of days (Andersen et al. 2016; Astor et al. 2013; Bouchard et al. 2012; Cohen et al. 2016; Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010). Three studies (18%) spanned for weeks (Dziembowska et al. 2016; Kim et al. 2012; Raaijmakers et al. 2013). The remaining seven studies (41%) ran over the course of months (Bradley et al. 2010; Henriques et al. 2011; Kudo et al. 2014; McCraty et al. 2009; Rusciano et al. 2017; Sherlin et al. 2013; Whited et al. 2014; Table 7).

4 Discussion

The results of this systematic review support a variety of physiological, psychological, and performance improvements when biofeedback interventions are administered for

stress management. One of the 17 total studies evaluated reported a lack of statistically significant change across all measures (Raaijmakers et al. 2013). The only significant findings in this study were two main effects of time, indicating an improvement in performance in the experimental group from pre- to post-intervention, as well as an improvement in performance for the control group from pre- to post-intervention. All remaining studies reported at least one meaningful significant difference in the experimental group following the intervention among either physiological, psychological, or performance assessments. In many cases, significant observations were made in more than one domain, and in some cases, in all three (Andersen et al. 2016; Bouchard et al. 2012; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2013).

4.1 Who benefits the most from biofeedback interventions?

When we take a closer look and consider the results in regards to the populations enrolled, we can make a more nuanced conclusion about specifically for whom a biofeedback intervention for stress management might benefit most. All eight studies that deliberately recruited specialized populations observed benefits across at least one measure within every domain analyzed. For example, all eight studies collected and analyzed physiological measures, and all eight reported improvement in the key components collected

Table 7 Intervention and study duration and frequency

References	Session	Frequency	Duration of study
McCraty et al. (2009)	15-min Heart Lock-Ins, 3 FreezeFrames per week; 30-s FreezeFrames/hour	Daily	3 months
Bradley et al. (2010)	TestEdge	2 times per week	5 months
Sherlin et al. (2010)	15 min	Once	1 day
Henriques et al. (2011)	15 min	5 times per week	1 month
Prinsloo et al. (2011, 2013a, b)	10 min	Once	1 day
Bouchard et al. (2012)	30 min	Daily	3 days
Kim et al. (2012)	15 min	3 total sessions	3 weeks
Astor et al. (2013)	25 min	Once	1 day
Sherlin et al. (2013)	20–35 min	Average of 3 sessions per week (15 total per participant)	1 month
Raaijmakers et al. (2013)	30 min	7 sessions	2 weeks
Escolano et al. (2014)	25 min	1 session	3 days
Kudo et al. (2014)	To a certain score (untimed)	At least 1 session per day	1 month
Whited et al. (2014)	32 min	4 to 8 sessions	1–3 months
Andersen and Gustafsberg (2016)	iPREP	No details	6 days
Cohen et al. (2016)	90–120 min	Once	1 day
Dziembowska et al. (2016)	20 min	10 sessions	3 weeks
Rusciano et al. (2017)	15–30 min	15 sessions	4 months

Provides details about the intervention frequency and study duration (iPREP: International Performance Resilience and Efficiency Program)

(Andersen et al. 2016; Bouchard et al. 2012; Dziembowska et al. 2016; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2013). Seven of the eight studies collected and analyzed psychological metrics, and all seven of those reported significant changes after the intervention (Andersen et al. 2016; Bouchard et al. 2012; Dziembowska et al. 2016; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2013). In terms of performance, all five studies that assessed performance also reported a significant improvement following the intervention (Andersen et al. 2016; Bouchard et al. 2012; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2013).

While biofeedback was also largely effective in the convenience samples, the results are less comprehensive. Eight of the nine studies collected and analyzed physiological data, and six of them (75%) reported a significant difference among at least one measure (Astor et al. 2013; Bradley et al. 2010; Escolano et al. 2014; Kim et al. 2012; Sherlin et al. 2010; Whited et al. 2014). Six of the nine studies also collected and reported psychological metrics, but only four total (67%) reported any significant differences (Bradley et al. 2010; Cohen et al. 2016; Henriques et al. 2011; Sherlin et al. 2010). Although seven of the nine studies collected and analyzed data representative of performance, only four of those (57%) reported significant differences (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Kim et al. 2012).

The difference observed here is stark, and has substantial implications for the application of biofeedback interventions. Inconsistencies among results in lab-based settings using convenience samples could imply that this setting is not truly stressful enough, or it could imply that individuals recruited through convenience sampling do not have high enough levels of stress to detect a difference following a stress management intervention. If a user's stress level is already quite low to begin with, this sort of intervention may not have as much value. However, when biofeedback is administered to individuals who operate under stressful conditions on a regular basis and who require appropriate stress management techniques to perform their job optimally, the intervention is extremely effective. Furthermore, this effectiveness is pervasive and can be observed across measures of physiology, subjective perceptions and cognitive state, as well as performance measures.

4.2 Consistency and standardization in data collection and measurement

Of equal importance when collecting and feeding back physiological data in near-real time is the equipment used to collect biopotential signals, and the software to process and transform that data in a sensitive and accurate

fashion (Laborde et al. 2017). There was very little overlap in regards to equipment used. To collect HRV and HR data, the Freeze-Framer/emWave and StressEraser devices were utilized the most, followed by chest-worn HR belts. Despite the frequent use of chest-worn HR belts, the specific models of belts were all unique. EEG data collection was also accomplished using a variety of devices. This has implications for the quality of data due to a broad variety of acquisition settings across different hardware. The rate at which this technology is advancing could make it difficult to standardize the specific equipment used in these studies, but to obtain transparency and reproducibility, the scientific community using these approaches should strive to at least use the same acquisition settings and detection thresholds for universal metrics, such as HRV and EEG.

The physiological measures themselves were largely comparable across studies, with many overlapping HRV components reported. There was, however, no overlap among EEG components reported as significantly different across studies. This may call into question the interpretations of EEG findings within this domain. Although none of the improvements in specific bandwidths were corroborated by other studies in this review, the absence of overlap observed could be the byproduct of a lack of standardized approaches, settings, experimental designs, and populations recruited. Because these differences existed among every study included, the same results cannot be expected to be replicated. Something more standard and specific, such as time-domain components of HRV (including SDNN, RMSSD, and pNN50), may be better served to make more generalized and universal claims.

Similarly, there was almost no consistency across psychological measures and the instruments administered to approximate those measures. Stress was a prominent and frequently evaluated construct, but the self-report instruments employed to measure stress varied widely. Overall, there were 23 unique scales or inventories used across the 14 studies that evaluated psychological constructs, eliminating the possibility of direct comparison of psychological assessment.

Beyond addressing physiological and psychological components of stress and stress management, a major focus of this review is the practical concern of how stress and stress management can influence task performance. For the purposes of this discussion, we will consider performance metrics and the experimental task insofar as they relate to the population evaluated. Of the seven studies recruiting convenience samples that also evaluated performance, five of those evaluated RT and/or errors made (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Raaijmakers et al. 2013; Sherlin et al. 2010), which speaks to relative consistency across studies. However, these metrics were assessed using a variety of tasks. Two studies enlisted the a version

of the Stroop task to evaluate performance (Escolano et al. 2014; Sherlin et al. 2010), while the remaining studies used either additional general cognitive batteries (Escolano et al. 2014; Raaijmakers et al. 2013) or tasks developed specifically for their research questions (Astor et al. 2013; Bradley et al. 2010; Cohen et al. 2016; Kim et al. 2012). Using consistent performance metrics contributes to greater comparability across convenience samples of students and healthy adults, but using the same task would ensure even greater comparability. Although four of these seven studies did report significant improvement according to at least one performance metric (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Kim et al. 2012), the variety across assessment tools used limits the generalizability of these results and reveals discrepancies in how performance is evaluated in the lab setting.

The studies evaluating performance within deliberately selected populations did use a wide variety of assessment approaches, but the majority of them were domain- and job-specific and, therefore, translatable to performance as a soldier (Bouchard et al. 2012), a professional baseball player (Sherlin et al. 2013), a police officer (Andersen and Gustafsborg 2016), and a elite professional soccer player (Rusciano et al. 2017). Performance among senior managers was evaluated using the Stroop Color-Word Test (Prinsloo et al. 2011, 2013a, b). It is worth noting the significant improvement in performance observed in all of these studies, despite the variety in populations recruited and assessment tools used. In fact, this speaks to the value in tailoring the context and the evaluation modality for performance among specialized populations.

4.3 Effects of intervention frequency and study duration on collected measures

The frequency of experimental sessions administering the biofeedback intervention was another major component that varied across studies. Most studies reported significant improvements within some measures across all types of data collected and analyzed, regardless of the duration of the study and the frequency of the intervention. For example, Escolano and colleagues collected and analyzed physiological and performance data, and reported at least one component within each of those types of data to have significantly changed in the experimental group following the intervention (Escolano et al. 2014). Notably, 80% of studies administering the intervention only once (Astor et al. 2013; Cohen et al. 2016; Escolano et al. 2014; Prinsloo et al. 2011, 2013a, b) saw improvements across all types of data collected and analyzed. Of the studies administering the intervention between 3 and 10 times, 60% likewise saw widespread improvements (Bouchard et al. 2012; Dziembowska et al. 2016; Kim et al. 2012). Finally 67% of those

administering the intervention more than 15 times (Kudo et al. 2014; McCraty et al. 2009; Rusciano et al. 2017; Sherlin et al. 2013) saw improvements across all types of data collected and analyzed. This supports the relative effectiveness of biofeedback as an intervention to address stress management, regardless of the frequency of intervention delivery.

When we consider the types of populations with the most widespread improvements in relation to the number of sessions administered, a more nuanced trend emerges. Overall, 25% of those administered in one session that saw widespread improvements were deliberately chosen samples (Prinsloo et al. 2011, 2013a, b); 67% of those administered between 3 and 10 times were deliberately chosen samples (Bouchard et al. 2012; Dziembowska et al. 2016); and 100% of those administered more than 15 times were deliberately chosen samples (Kudo et al. 2014; McCraty et al. 2009; Rusciano et al. 2017; Sherlin et al. 2013). This observation suggests that in more specialized populations, a more long-term intervention is associated with greater improvements across measures. Also worth noting is that the only studies reporting a lack of significant improvement in one or more types of data collected had recruited convenience samples (Bradley et al. 2010; Henriques et al. 2011; Raaijmakers et al. 2013; Sherlin et al. 2010; Whited et al. 2014), regardless of the number of sessions in which the biofeedback intervention was delivered.

4.4 Practical implications

Overall, these results seem to support the idea that contextualizing and personalizing a long-term biofeedback intervention for specialized populations would be the most efficacious approach. Despite the improvements observed across populations enrolled, assessments used, and frequency and duration of the intervention, it is clear that deliberately recruited populations experiencing high levels of stress benefit most when the intervention is equally deliberately designed. These claims cannot be made about convenience samples evaluated in the lab setting.

HRV biofeedback training has an established history of effectiveness across many populations and uses (Lehrer 2013a), as well as a promising future of additional uses including stress management (Lehrer 2007; Lehrer et al., 2008). Notably, throughout the studies reported in this review, HRV biofeedback was used to present the HRV waveform to users the most out of all delivery modalities. In practical terms, it is the most standardized approach with the strongest record of effectiveness, as well as the most profound mechanistic understandings (Eddie et al. 2015; Lehrer 2013b; Lehrer et al. 2013). Thus, visual biofeedback relying on a HRV waveform display and HRV biofeedback training principles could be a viable path forward to introduce greater

standardization in the field, while still affording the flexibility of tailoring variables such as frequency of intervention, duration of sessions, and performance measures to obtain results specific to a given population.

Though performance evaluations and measures should vary according to the population enrolled, standardization of measures should extend to physiological and psychological assessments as well. The State-Trait Anxiety Inventory for Adults was widely used throughout the studies reviewed here, and remains to be widely used for many studies evaluating stress. This instrument is also available as a 6-item, psychometrically validated short form (Marteau et al. 1992). Certain contexts and research questions would also require additional self-report measures, but maintaining one common measure would allow greater generalizability or comparison across the literature. In terms of physiological variables, EEG and HRV are typically considered to be sensitive, responsive, and robust indicators of stress states (Sharma et al. 2012). Other work has shown the positive association between taskload and cognitive workload, measured by HR, HRV, and respiratory rate, on a computer-based task, supportive of using advanced wearable sensors to detect stress and workload (Nixon et al. 2017). The research question of interest would likely determine specific EEG and HRV components analyzed (Laborde et al. 2017). Using these signals for analysis would thus ensure a level of reproducibility and comparability to past and future work, but would also allow the opportunity to adapt the analytical approach to the context at hand.

4.5 Additional findings and considerations

Our systematic review also revealed that the display of biofeedback was variable across studies. This is an important finding, because to optimize the use of biofeedback, it is necessary to have an appropriate display structure and integration with the task. The biofeedback modality used by 100% of researchers in this review was visual, and more than half of the displays (53%) presented physiological information as the HRV waveform (Fig. 2). Of the studies using HRV waveform as the primary visual component of the biofeedback intervention, 89% reported significant differences in physiology (Bradley et al. 2010; Dziembowska et al. 2016; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Rusciano et al. 2017; Sherlin et al. 2010; Whited et al. 2014) and 88% reported significant differences in psychological metrics (Bradley et al. 2010; Dziembowska et al. 2016; Henriques et al. 2011; Kudo et al. 2014; McCraty et al. 2009; Prinsloo et al. 2011, 2013a, b; Sherlin et al. 2010) following the intervention. Only 50% of those using HRV waveform information that evaluated performance detected significant differences in performance as a result of the intervention (Prinsloo et al. 2011, 2013a, b;

Rusciano et al. 2017). This difference among the percentage of change observed in performance compared to physiological and psychological measures could call into question the purported and previously observed curvilinear relationship between stress and performance (Allen 1983).

Aside from the HRV waveform display of information, the remaining modalities were completely unique. This is important to consider, because the method and display in which physiological changes are presented to the user could have a meaningful impact on the effectiveness of the intervention. Well-designed information displays can facilitate cognition and performance, but when designed poorly they can have the opposite effect, inducing cognitive decline and poorer performance (Kontogiannis et al. 1999). Essentially, for the intervention (a stimulus) to be effective, the human user must access and interpret it (the perception) readily, and act on it (the response) accordingly. To optimize this stimulus–response pattern, the stimulus presentation matters. Although visual cognition has a limited capacity, information presentation can complement and enhance that capacity (Alvarez 2011; Summerfield et al. 2009). Display components including location (Wickens et al. 1995), format (Atkins et al. 2002; Shah et al. 1995), icons (Talcott et al. 2007), and color (Wickens et al. 1990) of information objects all have an impact on information integration and decision-making, warranting a closer consideration of and approach to how information is presented. Recent work has confirmed that providing task-relevant information during cognitively demanding tasks can improve performance and problem-solving capacities (Dadashi et al. 2017). If a major component of the task is the ability to self-regulate and manage stress levels, physiological information displayed through a biofeedback modality could be equally valuable as performance feedback.

4.6 Conclusion

The findings from this systematic review reveal that biofeedback interventions for stress management have positive effects, and are particularly effective among deliberately selected populations. A biofeedback intervention, which uses physiological output in an effort to mediate real-time internal states, could potentially aid individuals by regaining an allostatic balance conducive to functional cognition and performance. Considering the neurovisceral integration model (Thayer et al. 2009, 2012), results from this review support that restoring one's physiological state by focusing on HRV, which was demonstrated to be the most common form of self-regulation training, could affect cognitive resources in a way that enhances performance relying on executive functioning and attention processing. This effect is especially pronounced in high-risk professions. If we consider the curvilinear relationship between stress and

performance (Fig. 1) that has been proposed and supported in the literature, this conclusion can be readily understood. Individuals working in high-risk professions are likely operating in the realm of distress, sometimes past the point of optimal stress. Incorporating biofeedback as a stress management approach is an effective way to reposition oneself in the realm of eustress via behavioral adaptation. Support for this theory was not found within studies recruiting convenience samples, indicative that this population, mostly comprised of college students and healthy adults, may operate at a lower level of stress that experimental stressors cannot as easily elevate. In this case, biofeedback would reasonably be less effective, because there is a lower stress level to manage at the outset.

The pool of literature acquired had some limitations. The relatively limited number of empirical studies that met our inclusion criteria points to a dearth in research specifically on using biofeedback for the purposes of stress management. The potential implications of using biofeedback as a stress management tool are substantial, but the broad variety in approaches and in results are indicative of the limited consistency across studies. Yet based on the results, of studies recruiting specialized populations, stress reduction and performance enhancement may be more likely to be achieved when the task and metrics are specialized according to the population. With the overall intent to use biofeedback as a training intervention in these populations specifically, users would ideally learn to maintain a healthy physiological balance without continued reliance on the biofeedback device in the long term (Schwartz 2010). Longitudinal follow-up is thus imperative to confirm the long-lasting effects of biofeedback we wish to instill, yet only one of the 17 studies included reported follow-up data of 90 days post-intervention (McCraty et al. 2009), a relatively short period to observe a long-term effect. Their own conclusions emphasize the need for a longer follow-up period, citing that chronically stressed individuals, such as those in high-risk professions, require 6–9 months for full recovery, especially considering hormonal stress markers such as cortisol and dehydroepiandrosterone, evaluated in their study (McCraty et al. 2009). Only one other study measured cortisol (Bouchard et al. 2012), but authors make no mention of following up beyond the duration of the study.

Despite the limitations, results of this systematic review are promising. While the acute stress response can be

adaptive and appropriate in certain contexts, its inappropriate management and continuous exposure has proven to be detrimental to high-risk practitioners (Arora et al. 2010a, b; Joseph et al. 2016; LeBlanc 2009; Mazur et al. 2013, 2014; Moorthy et al. 2003; Pluyter et al. 2010; Weigl et al. 2016; Wetzel et al. 2006; Yurko et al. 2010), who are prone to experience frequent exposures to acute and chronic stressors with little recovery time available. The issue of chronic stress and its effects on health, quality of life, and professional performance is also of primary concern (Balch et al. 2009; Rosenstein 2012). When stress levels are maintained within an optimal range, however, growth and/or positive performance are facilitated (Allen 1983; Vine et al. 2013). Ensuring optimal perceptions of stress among practitioner professions is integral, and this population has a substantial potential to benefit from biofeedback administration. The findings from this review illustrate that the systematic evaluation, application, and influence of such technology among non-patients, however, is understudied. Collectively, the results from these studies convey beneficial outcomes across physiological, psychological, and performance measures in various high-performing and acutely and chronically stressed populations, but the limited number of studies reviewed point to a need for more widespread implementation and evaluation. The diversity in methods also suggests a potential benefit for standardization among general populations, and specialization among specific populations.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Research involving animal and human participants This article does not contain any studies with human participants or animals performed by any of the authors.

Appendix A

References	Study population	Sample (Expr/Ctrl)	Mean age (stdev)	Modality: visual	Modality: auditory	Physio measures	Physio equipment	Physio results
Evetovich et al. (2007)	College students	12	22 (1.1)	MMG signal	N/A	MMG, EMG	BioPac	Expr vs. Ctrl: ↓ MMG amplitude, EMG amplitude
McCraty et al. (2009)	Correctional officers	75 (43/32)	Expr: 39.47 (7.70) Ctrl: 40.72 (8.12)	HRV waveform (coherence)	N/A	Cortisol, DHEA, S-IgA cholesterol, triglycerides, fasting glucose levels, HR, HRV, BP	Freeze-Framer/emWave	Pre-post Expr: ↓ total cholesterol, LDL, total/HDL ratio, glucose levels, DHEA, mean arterial pressure, systolic BP, diastolic BP, mean HR, ↑ LF/HF ratio of HRV; Pre-post Ctrl: ↓ total cholesterol, HDL, glucose levels
Bradley et al. (2010)	High school students	136 (77/59)	Expr: 15.3 (0.44) Ctrl: 15.3 (0.44)	HRV waveform (coherence)	N/A	HRV	Freeze-Framer/emWave	Pre-post Expr vs. Ctrl: ↓ HR, ↑ SDNN, HF, LF, IC
Sherlin et al. (2010)	Healthy adults	43	33.2 (8.77)	HRV waveform (coherence)	N/A	QEEG	StressEraser	Pre-post Expr: ↑ overall alpha; Pre-post Ctrl: ↓ overall alpha
Henriques et al. (2011)	College students	35	Not reported	HRV waveform (coherence)	N/A	HRV	emWave	Pre-post Expr: no significant differences
Prinsloo et al. (2011, 2013a, b)	Senior managers	18 (9/9)	Expr: 33 (6) Ctrl: 34 (6)	HRV waveform (coherence)	N/A	BP, EEG, HR, HRV, RR, RF	BioPac, StressEraser	Interv. Expr vs. Ctrl: ↓ RR, RF, ↑ LF power, TF power, (2011), ↓ relative frontal and central beta, ↑ SDNN, relative frontal theta, frontal, central, and parietal theta/beta ratio (2013b) Pre-post Expr: ↓ HR (2011); Pre-post Ctrl: ↓ HR (2011)
Bouchard et al. (2012)	Soliders	41 (21/20)	24.9 (5.55)	Red texture partially obscuring user's field of view. Increased arousal represented by an increasingly obstructed view	Sound of a pumping heart increasing in frequency and loudness with increased arousal	Daily salivary cortisol, HR	Salivette, Polar wireless transmitter belt	Post Expr vs. Ctrl: ↓ cortisol; Mid Expr vs. Ctrl: ↓ HR
Kim et al. (2012)	High school students	46 (21/25)	Not reported	ECG tachogram, anxiety profile	N/A	HRV	QECG-3	Pre-post Expr: ↑ R-R intervals

References	Study population	Sample (Expr/Ctrl)	Mean age (stdev)	Modality: visual	Modality: auditory	Physio measures	Physio equipment	Physio results
Astor et al. (2013)	College students	Study 1: 36 (19/17); Study 2: 68 (44/24)	Study 1: 23.39 (range: 8); Study 2: 22.06 (range: 9)	Color-coded arousal meter; corresponding game elements changing color to reflect the state of arousal, located directly in attentional field	N/A	HR (arousal level)	Chest-worn HR belt	Study 1 Expr vs. Ctrl: not reported; Study 2 Expr vs. Ctrl: ↓ arousal level
Sherlin et al. (2013)	Professional male baseball players	5	19.2 (0.45)	Dual Drive Xtreme: car presented in a race, moves at a speed proportional to user success; OR Particle Editor: shapes or lines in movement along with music when parameters are met	Music	QEEG	Electro-Cap International	Pre-post: ↑ central beta (4/5), frontal beta (3/5), parietal beta (3/5)
Raaijmakers et al. (2013)	College students	28 (16/12)	22 (range: 8)	Across 6 tasks: 3 displayed SCL signal, 2 displayed HRV signal, 1 displayed a combination of HRV and SCL signals	N/A	HRV, SCL, EEG alpha asymmetry	ActiveTwo	Pre-post Expr vs. Ctrl: No significant differences
Escolano et al. (2014)	College students	19 (10/9)	Expr: 25.8 (4.1) Ctrl: 24.3 (3.7)	A square with changing saturation colors; physiological values reflecting baseline appear gray, values above baseline appear red and increase in saturation as values increase, values below baseline appear blue and increase in saturation as values decrease	N/A	EEG	Guger Technologies	Pre-post Expr: ↑ UA power, LA2

References	Study population	Sample (Expr/Ctrl)	Mean age (stdev)	Modality: visual	Modality: auditory	Physio measures	Physio equipment	Physio results
Kudo et al. (2014)	Post-partum mothers	55 (25/30)	Expr: 30.5 (5.7) Ctrl: 33.4 (6.6)	HRV waveform (coherence)	N/A	HRV	StressEraser	Pre-post Expr vs. Ctrl: ↓ HR, ↑ SDNN, HF power
Whited et al. (2014)	Under-graduate and graduate students	28 (15/13)	Expr: 22.29 (3.6) Ctrl: 23.15 (4.06)	HR, HRV power spectrum, HRV waveform (coherence)	N/A	HR, HRV	Polar RS800CS, emWave	Pre-post Expr vs. Ctrl: ↑ pNN50 stress reactivity
van Dijk et al. (2015)	Healthy adults	74	27 (range: 49)	Trending HR over last 3 min, current HR in BPM	N/A	HR	Kendall H124SG	Expr_HR vs. Ctrl: ↑ correlation between HR and momentary/retrospective stress
Zauszniewski et al. (2015)	Grand-mothers raising grand-children	20	58 (range: 26)	HRV waveform (coherence)	N/A	HRV	StressEraser	(-) correlation between perceived stress and coherence, negative emotions and coherence, and depressive cognitions and coherence
Andersen et al. (2016)	Police officers	12 (6/6)	31.5 (range: 7)	No details	N/A	HR, BP	Zephyr BioHarness, Omnisense ambulatory blood pressure wrist cuffs	Post Expr vs. Ctrl: ↓ HR max between baseline and Scenario 1, ↓ change in HR from baseline to Scenario 1
Cohen et al. (2016)	Healthy adults	29	25.5 (4.67)	Performance prediction bar graph, HR bar graph, error chance prediction bar graphs	N/A	HR, HRV	Zephyr HxM	Not reported
Dziembowska et al. (2016)	Young male athletes	41 (20/21)	18.34 (1.36)	HRV waveform (coherence)	N/A	EEG alpha asymmetry, HRV	emWave, Electro-Cap International	Pre-post Expr: ↑ HRV spectral power, LF HRV, IC, theta and alpha spectral power, alpha asymmetry, ↓ HF HRV
Rusciano et al. (2017)	Professional male soccer players	20 (10/10)	Expr: 30.0 (3.8) Ctrl: 30.7 (4.3)	Sessions 1–3: HRV waveform (coherence); Sessions 4–9: HRV waveform (coherence), SCL level, EMG signal, skin temperature signal; Sessions 10–15: HRV waveform (coherence)	N/A	SCL, RR, BVP, HR, HRV, surface EMG, skin temperature	Nexus 10 Mark II	Post Expr vs. Ctrl: ↓ SCL, HR, RR; Pre-post Expr: ↓ RR ↑ LF

References	Psych measures	Psych instruments	Psych results	Performance measures	Performance instruments	Performance results	Session	Frequency	Duration of study	Notes
Evetovich et al. (2007)	N/A	N/A	N/A	Number of repetitions	Seated preacher curl exercise	Expr vs. Ctrl: No significant differences	Until fatigued	1 session every 2 days (5 total per participant)	9 days	Eliminated: physical stress
McCraty et al. (2009)	Health risk, Type A and coronary-prone behavior, psychological symptoms, psychological and workplace elements	Personal Wellness Profile, Jenkins Activity Survey, Brief Symptom Inventory, Personal and Organizational Quality Assessment	Pre-post Expr: ↓ anger, fatigue, hostility, interpersonal sensitivity, paranoid ideation, psychological distress, Type A behavior, speed and impatience, ↑ productivity, motivation, goal clarity, perceived manager support, gratitude, positive outlook; Pre-post Ctrl: ↓ work attitude, confidence in organization, ↑ depression	N/A	N/A	N/A	15-min Heart Lock-Ins, 3 Freeze-Frames per week; 30-s Freeze-Frames/hr	Daily	3 months	
Bradley et al. (2010)	Test anxiety (global, worry, emotionality)	Test Anxiety Inventory (8 of 16 items)	Pre-post Expr vs. Ctrl: ↓ overall test anxiety (global, worry, emotionality)	Standardized testing scores	California High School Exit Exam, California Standards Test	Pre-post Expr vs. Ctrl: No significant differences	TestEdge	2 times per week	5 months	
Sherlin et al. (2010)	State anxiety	State-Trait Anxiety Interview-State Form	Pre-post Expr vs. Ctrl: ↓ state anxiety	Incorrect vs. correct response, reaction time, missed responses	Modified Stroop task	Pre-post Expr vs. Ctrl: No significant differences	15 min	Once	1 day	

References	Psych measures	Psych instruments	Psych results	Performance measures	Performance instruments	Performance results	Session	Frequency	Duration of study	Notes
Henriques et al. (2011)	Mood/ anxiety symptoms, state and trait anxiety, subjective well-being	Mood and Anxiety Symptom Questionnaire, State-Trait Anxiety Inventory, Scales of Psychological Well-Being	Pre-post Expr: ↓ general distress: mixed, general distress: anxious, anxious arousal	N/A	N/A	N/A	15 min	5 times per week	1 month	
Prinsloo et al. (2011, 2013a, b)	State and trait anxiety, relaxation, sleepiness	State-Trait Anxiety Inventory Forms Y-1 and Y-2, Smith Relaxation States Inventory 3, Visual Analog Scale	Pre-post Expr vs. Ctrl: ↑ relaxation, ↓ sleepiness	Reaction time, mistakes made	Stroop task	Pre-post Expr vs. Ctrl: ↑ reaction time, consistency in responses, ↓ mistakes made	10 min	Once	1 day	
Bouchard et al. (2012)	Perceived self-efficacy to control stress	Perceived self-efficacy, Simulator Sickness Questionnaire, appreciation and usefulness of program	Pre-post Expr vs. Ctrl: ↑ perceived self-efficacy to control stress	Critical task completion, program elements met	Trainees Evaluation Sheet, ImPACT coach checklist	Post Expr vs. Ctrl: ↑ performance	30 min	Daily	3 days	
Kim et al. (2012)	N/A	N/A	N/A	Speech performance	Korean Broadcasting System speech performance test	Pre-post Expr: ↑ speech performance	15 min	3 total sessions	3 weeks	
Astor et al. (2013)	Cognitive reappraisal and suppression	Emotion Regulation Questionnaire	Not reported	Percentage of correct decisions	Auction Game decision performance	Study 1 Expr vs. Ctrl: ↑ performance; Study 2 Expr vs. Ctrl: ↑ performance	25 min	Once	1 day	
Sherlin et al. (2013)	Focus index, stress index, speed index	NeuroPerformance Profile	Pre-post: ↑ sleep, focus, feeling of being relaxed, all indices of NeuroPerformance Profile	Response time, response time variability, errors of omission, errors of commission	QIKtest continuous performance test	Pre-post: ↓ errors of commission	20–35 min	Average of 3 sessions per week (15 total per participant)	1 month	

References	Psych measures	Psych instruments	Psych results	Performance measures	Performance instruments	Performance results	Session	Frequency	Duration of study	Notes
Raaijmakers et al. (2013)	Affect	Positive and Negative Affect Scale	Pre–post Expr vs. Ctrl: No significant differences	Reaction time, error percentages	N-back task, Mental Rotation Task	Pre–post Expr: ↓ RT, ↑ accuracy Pre–post Ctrl: ↓ RT, ↑ accuracy	30 min	7 sessions	2 weeks	
Escolano et al. (2014)	N/A	N/A	N/A	Errors, elapsed time, recognized words, interference, correct responses, reaction time	Paced Auditory Serial Addition Task (PASAT), Rey Auditory Verbal Learning Test (RAVLT), Trail Making Test (TMT), Stroop Color-Word Test, Mental Rotation Task (MRT)	Pre–post Expr vs. Ctrl: ↑ part B of TMT; Pre–post Expr: ↓ errors and time elapsed in PASAT, reaction time in MRT, ↑ recognized words in RAVLT, correct responses in MRT; Pre–post Ctrl: ↓ time elapsed in PASAT, ↑ correct responses in MRT, part A of TMT	25 min	1 session	3 days	
Kudo et al. (2014)	Postnatal depression	Edinburgh Postnatal Depression Score	Pre–post Expr vs. Ctrl: ↓ EPDS score, anxiety, difficulty sleeping, sad and miserable thoughts	N/A	N/A	N/A	To a certain score (untimed)	At least 1 session per day	1 month	
Whited et al. (2014)	Distress, stress, state affect	Brief Symptom Inventory, Perceived Stress Scale, Multiple Adjective Affect Checklist-Revised	Pre–post Expr vs. Ctrl: No significant differences	N/A	N/A	N/A	32 min	4 to 8 sessions	1–3 months	

References	Psych measures	Psych instruments	Psych results	Performance measures	Performance instruments	Performance results	Session	Frequency	Duration of study	Notes
van Dijk et al. (2015)	Momentary stress, retrospective stress, neuroticism, anxiety sensitivity	Likert-type scales, Big 5 Personality Inventory, Anxiety Sensitivity Index	Ctrl: (+) correlation between neuroticism and momentary/retrospective stress; Expr: Stress: (+) correlation between anxiety sensitivity and momentary/retrospective stress	N/A	N/A	N/A	20 min	Once	1 day	Eliminated: correlational analyses only
Zauszniewski et al. (2015)	Perceived stress, negative emotions, depressive cognitions	Perceived Stress Scale, Emotional Symptom Checklist, Depressive Cognition Scale	(-) correlation between perceived stress and coherence, negative emotions and coherence, and depressive cognitions and coherence	N/A	N/A	N/A	10–15 min	At least 28 sessions	3 months	Eliminated: correlational analyses only
Andersen et al. (2016)	Perceived stress, skills, ability to manage scenarios, self-confidence in decision-making, situational awareness	Stress, Skills, Ability to manage scenarios, Situational awareness	Pre Ctrl vs. Expr: ↑ confidence	Critical task completion, situational awareness, decision-making, overall performance	Senior trainers' ratings	Pre-post Expr vs. Ctrl: ↑ critical task completion; Post Expr vs. Ctrl: ↑ overall performance, situational awareness, decision-making	International performance resilience and efficiency program (iPREP)	No details	6 days	
Cohen et al. (2016)	Appraisal, task demand, usability	Likert-type scale, Level-of-Information Processing Scale, System Usability Scale	HR vs. noHR: ↓ perceived usability of the feedback	Total score for task, number of errors made	Simulated fire management task	Expr vs. Ctrl: ↑ relative performance score	90–120 min	1 session	1 day	

References	Psych measures	Psych instruments	Psych results	Performance measures	Performance instruments	Performance results	Session	Frequency	Duration of study	Notes
Dziembowska et al. (2016)	State anxiety, self esteem	State-Trait Anxiety Inventory A-State Scale, Rosenberg Self Esteem Scale	Pre-post Expr: ↓ state anxiety	N/A	N/A	N/A	20 min	10 sessions	3 weeks	
Rusciano et al. (2017)	N/A	N/A	N/A	Accuracy, reaction time, injury prevention	Stroop task, visual search task	Pre-post Expr vs. Ctrl: ↑ accuracy in Stroop task, presence at training days ↓ RT in visual search task, absences at training days, injuries	15–30 min	15 sessions	4 months	

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